The rise of the flying firehose robot

Scientists at Osaka University in Japan have introduced the Dragon Firefighter, a flying firehose prioritising the safety of human firefighters. This remotely controlled prototype, measuring four metres long, is designed to approach building fires directly, ensuring efficient and secure extinguishing procedures. It suspends the firehose approximately two metres above the ground, employing eight adjustable water jets for propulsion, granting it flexibility in shape and direction and facilitating precise targeting of flames. It operates through a unit in a trailing wheeled cart connected to a fire truck, featuring a 14,000-litre water reservoir. Recent enhancements include improved waterproofing, a versatile nozzle unit, and an efficient water flow control mechanism.

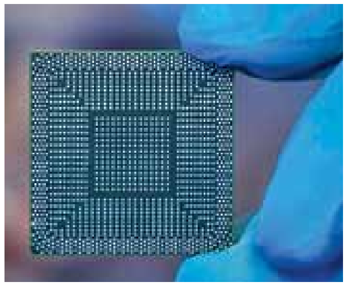
Flying firehose robot (Credit: https://dronelife.com)

Real-time road alerts with 360-degree display

Researchers from the Universities of Cambridge, Oxford, and UCL have advanced vehicle heads-up display systems using 3D laser scanning and lidar data to create a comprehensive three-dimensional representation of London's streets. Unlike traditional two-dimensional projections, their system can project lifelike holographic representations of concealed road obstacles, aligning them with the actual size and distance of real objects. This breakthrough technology enhances driver safety by dynamically projecting road hazards and obstacles from multiple angles in real time, directing the driver's focus towards the road. The team's augmented reality system combines 3D holography with lidar data, providing a 360° view with depth perception, and they are collaborating with Google for real-world car testing, scheduled for 2024.

Wireless under-skin chargers for medical implants

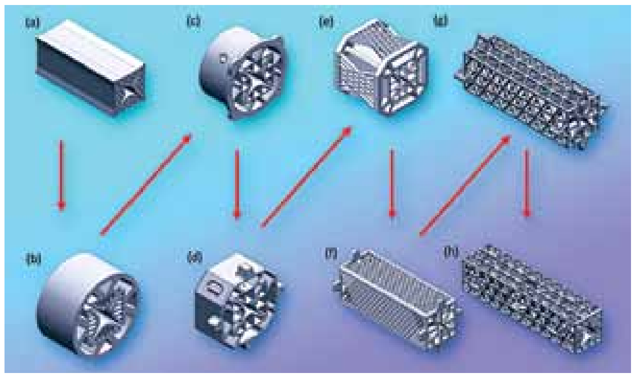
Researchers at Lanzhou University in China have created a wireless charging unit designed for placement beneath the skin to power internal bioelectronic devices. Successful rodent tests have demonstrated the potential to transform medical implants by eliminating the need for bulky batteries and intricate wiring. Scientists have devised flexible, biodegradable under-skin chips that conform to tissue contours during surgical procedures to tackle this issue. They integrated their prototype into a biodegradable chip-like implant, combining energy harvesting and storage capabilities. When connected to a medical device, this implant supplies consistent power by channelling energy through the circuit into a capacitor. During rat experiments, the wireless implant efficiently functioned for up to 10 days and fully biodegraded within two months, confirming its biodegradability.



A wireless charger that sits under the skin could power medical devices before dissolving into the body (Credit: www-wionews-com.cdn.ampproject.org)

3D printing of portable mass spectrometer components

MIT researchers have revolutionised mass spectrometry by creating a lightweight, cost-effective mass filter. This 3D-printed component is made from robust, heat-resistant glass-ceramic resin, eliminating the need for assembly and potential defects. The additive manufacturing process produces miniaturised quadrupoles with optimal size and shape for enhanced precision and sensitivity. The glass-ceramic resin can withstand high temperatures. Through vat photopolymerisation, the device is built layer by layer as a piston moves, featuring hyperbolic rods that are challenging to produce conventionally. A supportive triangular lattice structure ensures stability and electroless plating coats the rods in a conductive metal film, maintaining device integrity. These 3D-printed filters rival larger, commercial counterparts due to their precision and low density.



Because additive manufacturing allows the researchers to easily try new designs, they created a number of different quadrupole filters before arriving at the final iteration (h), which is surrounded by a series of triangular lattices to provide durability (Credit: Courtesy of the researchers, edited by MIT News)